

Sabaragamuwa University of Sri Lanka



REAL-TIME FACIAL EMOTION RECOGNITION USING MEDIAPIPE AND DEEP CONVOLUTIONAL NEURAL NETWORKS WITH FOCAL LOSS

A Project Report submitted to the Faculty of Computing, Sabaragamuwa University of Sri Lanka, in partial fulfilment of the requirements for the Degree of Bachelor of Science Honours in Data Science.

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Appendix A – Declaration

I declare that this report does not incorporate, without acknowledgment, any material previously submitted for a Degree or a Diploma in any University, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person or ourself except where due reference is made in the text. Also, I hereby grant Sabaragamuwa University of Sri Lanka the non-exclusive right to reproduce and distribute my report, in whole or in part, in print, electronic, or other media. I retain the right to use this content in whole or in part in future works (such as articles or books).

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Appendix B – Certificate of Approval

We hereby declare that this report is the student's own work and effort, and that all other sources of information used have been acknowledged. This report has been submitted with our approval.

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I am grateful to my fellow students in the Department of Data Science for their collegial support and stimulating technical discussions throughout the programme. Finally, I acknowledge the unrelenting patience and encouragement of my family, whose support sustained my motivation throughout the demanding period of research and report preparation.

Abstract

Automated facial emotion recognition (FER) in unconstrained, real-world environments presents a persistent challenge in affective computing, largely due to intra-class appearance variance, inter-class expression ambiguity, and severe dataset class imbalance. This capstone project addresses these challenges by designing, training, and deploying an end-to-end real-time FER system capable of classifying live webcam frames into seven emotion categories: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

The proposed methodology employs Google MediaPipe BlazeFace [3] for face bounding box localisation, and a purpose-built four-stage Deep Convolutional Neural Network (CNN) incorporating Batch Normalisation [7], L2 regularisation, Global Average Pooling, and progressive Dropout. To address the severe class imbalance inherent in the FER2013 benchmark [1] — where the Disgust class contains only 436 training samples compared to 7,215 Happy samples (a ratio of $16.5\times$) — a Categorical Focal Loss formulation ($\gamma = 2.0$) is combined with Label Smoothing ($\epsilon = 0.1$) and dynamic class weighting.

Empirical evaluation on 7,178 unseen FER2013 test images demonstrates that the proposed system achieves a test accuracy of 63.51% and a weighted F1-score of 62.91%, closely approaching the human annotation baseline of approximately 65% reported by Goodfellow et al. [1]. A 10-frame temporal prediction window based on majority voting substantially reduces high-frequency prediction instability during live deployment. An ablation study confirms that each proposed component — Focal Loss, Label Smoothing, class weighting, and the four-stage depth — contributes incrementally to the final performance.

The system operates at approximately 28–30 frames per second on consumer-grade hardware. Limitations regarding dataset generalisation, demographic bias, and ethical deployment considerations are discussed. Future directions include face-specific transfer learning encoders, Vision Transformer architectures, and multimodal emotion fusion.

Keywords: Facial Emotion Recognition, Convolutional Neural Network, Focal Loss, Label Smoothing, MediaPipe, Class Imbalance, FER2013, Real-Time.

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Glossary of Abbreviations

Abbreviation	Expansion and Definition
AU	Action Unit — a discrete facial muscle movement defined by the Facial Action Coding System (FACS)
BN	Batch Normalisation — a layer that normalises activations across mini-batch dimensions
CCE	Categorical Cross-Entropy — standard multi-class classification loss function
CNN	Convolutional Neural Network — a feed-forward DNN using learnable convolutional filters
DNN	Deep Neural Network — an artificial neural network with multiple hidden layers
FER	Facial Emotion Recognition — automated classification of human facial expressions
FER2013	Facial Expression Recognition 2013 — a publicly available benchmark dataset
FL	Focal Loss — a loss function that downweights easy examples during training
FPS	Frames Per Second — real-time video throughput metric
GAP	Global Average Pooling — spatial aggregation of feature maps into a vector
GPU	Graphics Processing Unit — parallel processor used for accelerated DNN training
HOG	Histogram of Oriented Gradients — a classical image feature descriptor
LBP	Local Binary Pattern — a texture-based hand-crafted feature descriptor
LS	Label Smoothing — regularisation technique that softens hard one-hot targets
ML	Machine Learning — automated learning of patterns from data
ROI	Region of Interest — cropped face bounding box region

SUSL	Sabaragamuwa University of Sri Lanka
SVM	Support Vector Machine — classical discriminative classifier
TF	TensorFlow — Google's open-source ML framework
UML	Unified Modelling Language — standardised notation for software design
ViT	Vision Transformer — transformer architecture applied to image patches
WSL	Windows Subsystem for Linux — Linux environment on Windows

Chapter 1: Introduction

1.1 Background and Context

Facial expressions are among the most immediate and universally recognisable channels of human non-verbal communication [4]. The ability to automatically interpret emotional states from facial imagery has broad and transformative implications across multiple application domains, including adaptive e-learning systems that modify instructional pacing based on student engagement, mental health diagnostic tools for tele-psychiatry, driver monitoring systems for drowsiness and fatigue detection, customer experience analytics in retail environments, and social robotics for empathetic human-robot interaction.

The interdisciplinary field of Affective Computing, first formalised by Picard [11], encompasses computational methods that recognise, interpret, and simulate human emotional states. Within this domain, Facial Emotion Recognition (FER) using deep learning has emerged as the predominant technical approach, largely superseding classical hand-crafted feature methods since the publication of the FER2013 benchmark dataset and deep learning results by Goodfellow et al. in 2013 [1].

Despite significant research progress, deploying FER systems in real-world scenarios remains technically challenging. Three compounding factors contribute to this complexity: (1) the high intra-class variability of facial expressions across different individuals and cultural backgrounds; (2) the inter-class ambiguity between visually similar emotions such as Fear and Surprise; and (3) the severe class imbalance inherent in publicly available FER datasets, particularly FER2013, where the least-represented emotion class (Disgust) contains approximately 16.5 times fewer training samples than the most prevalent class (Happy) [1].

1.2 Problem Statement

Standard deep learning classifiers trained with Categorical Cross-Entropy (CCE) loss on severely imbalanced datasets produce classification boundaries systematically biased towards majority classes. This results in high aggregate accuracy but systematically poor recall for minority emotion classes (Fear, Disgust, Surprise), which are frequently the emotionally significant classes in clinical and safety-critical applications. Furthermore, frame-level emotion predictions from neural networks are inherently noisy and unstable in live video due to transient facial micro-expressions, partial occlusions, and minor head movements, producing distracting prediction flickering in deployed systems.

Existing solutions have addressed class imbalance through oversampling (SMOTE) or modified loss functions, and have addressed detection robustness through Haar Cascade

classifiers or MTCNN detectors. However, few published systems integrate Focal Loss with Label Smoothing, dynamic class weighting, and MediaPipe-based face detection within a single, end-to-end real-time deployable pipeline. This project addresses this integration gap.

1.3 Goals and Objectives

The overarching goal of this capstone project is to design, implement, evaluate, and deploy a complete real-time FER system on consumer-grade hardware that addresses the class imbalance and prediction stability challenges identified above. The specific measurable objectives are:

- O1 — Design and train an upgraded 4-Stage Deep Convolutional Neural Network capable of classifying the seven FER2013 emotion categories.
- O2 — Formulate and implement a Categorical Focal Loss function ($\gamma=2.0$) combined with Label Smoothing ($\epsilon=0.1$) to improve minority-class recall.
- O3 — Compute and apply dynamic class weights to further amplify the gradient contribution of underrepresented emotion classes during training.
- O4 — Integrate Google MediaPipe BlazeFace [3] for frame-level face detection under variable illumination and moderate head pose variation.
- O5 — Implement a 10-frame temporal majority-vote prediction window to substantially reduce high-frequency prediction instability in live deployment.
- O6 — Conduct rigorous evaluation on the full FER2013 test set (7,178 images), analysing per-class accuracy, precision, recall, F1-score, and confusion matrix structure.
- O7 — Conduct an ablation study to quantify the incremental contribution of each proposed component to overall system performance.

1.4 Motivation

The motivation for this project arises from two complementary observations. First, from a technical standpoint, the persistent class imbalance problem in FER2013 has not been fully resolved by existing published solutions. The Focal Loss formulation of Lin et al. [2], originally proposed for dense object detection, has demonstrated significant potential for imbalanced classification tasks but remains under-explored in the FER domain in combination with Label Smoothing and dynamic class weighting. This project provides a systematic ablation study to quantify its effectiveness in this specific context.

Second, from a deployment standpoint, most published FER research reports offline evaluation on static test sets but does not address the engineering challenges of live webcam deployment: face detection stability, inference speed, and temporal prediction smoothing. This project bridges the gap between research benchmark evaluation and practical real-time deployment, producing a deployable single-click Windows application.

1.5 Scope of the Completed Project

The scope of this project is intentionally bounded to ensure feasibility within the capstone timeline. Specifically:

- The FER2013 dataset is the sole training data source; no additional proprietary datasets are incorporated.
- The system classifies a single primary detected face per frame; multi-face simultaneous classification is not supported in this version.
- The deployment environment targets Windows 10/11 consumer hardware with integrated webcam or USB camera input.
- Model architecture is a custom CNN; pre-trained deep architectures (VGGFace2, AffectNet) are evaluated conceptually in future work but not empirically implemented.
- Evaluation is conducted exclusively on the FER2013 official test partition; cross-dataset generalisation evaluation is a direction for future work.

1.6 Approach and Assumptions

The technical approach assumes that input video is captured at 720p resolution at 30 FPS via OpenCV VideoCapture, and that at least one primary human face is visible and frontally oriented within approximately $\pm 45^\circ$ of the camera axis. Face detection is fully delegated to MediaPipe BlazeFace [3], which has demonstrated sub-5ms face detection latency on CPU-class hardware. The CNN classifier operates on 48×48 pixel normalised grayscale tensors, consistent with the FER2013 image format.

A key engineering assumption is that the emotional expression displayed by the user is relatively sustained (>300 ms duration), allowing the 10-frame temporal window to produce stable majority-vote labels without requiring explicit expression onset/offset segmentation.

1.7 Research Contributions

The primary technical contributions of this project are:

- A systematically designed 4-Stage Deep ConvNet with double convolutional layers per block, Batch Normalisation, progressive Dropout, and Global Average Pooling, achieving 63.51% test accuracy on FER2013.
- A Categorical Focal Loss implementation with integrated Label Smoothing, formulated to match the asymmetric imbalance characteristics of FER2013.
- A structured ablation study quantifying the marginal performance contribution of six system configuration variants.
- An end-to-end real-time deployment pipeline integrating MediaPipe face detection with a temporal prediction window for consumer hardware deployment.
- Comprehensive documentation of system limitations, ethical considerations, and a structured real-time testing protocol.

1.8 Report Organisation

The remainder of this report is structured as follows. Chapter 2 presents a critical review of the literature on FER, deep learning architectures, loss function design, and class imbalance mitigation. Chapter 3 provides formal system specifications, use case modelling, and architectural diagrams. Chapter 4 details the complete methodology, from data preprocessing through training configuration and real-time pipeline design. Chapter 5 presents and discusses experimental results including ablation study, SOTA comparison, per-class metrics, and real-time testing results. Chapter 6 addresses technical limitations and ethical considerations. Chapters 7 and 8 present future work and conclusions respectively.

Chapter 2: Literature Review

2.1 Introduction to Facial Emotion Recognition

Facial Emotion Recognition (FER) is the automated process of detecting, analysing, and classifying human facial expressions into discrete emotional states. The theoretical foundation of FER research draws on Ekman and Friesen's [12] cross-cultural study of universal facial expressions, which proposed that six primary emotions — Anger, Disgust, Fear, Happiness, Sadness, and Surprise — are universally expressed and recognised across cultures. Subsequent researchers have debated this universality, particularly regarding cultural display rules and contextual ambiguity, but these six categories (plus Neutral) remain the dominant classification schema in computational FER research.

The Facial Action Coding System (FACS), also developed by Ekman and Friesen, provides a fine-grained anatomical description of facial expressions in terms of 44 Action Units (AUs) corresponding to specific facial muscle contractions. Many modern FER systems implicitly learn to detect combinations of AUs through convolutional feature hierarchies, even without explicit AU supervision.

2.2 Classical Feature-Based Methods

Prior to the deep learning era, FER systems relied on carefully engineered feature extraction pipelines. The most prevalent classical approaches include:

- Histogram of Oriented Gradients (HOG): Shan et al. [cited in: related literature] demonstrated that HOG features capture facial texture gradients and local shape information, achieving reasonable accuracy on constrained FER datasets when combined with linear SVMs. However, HOG features are not invariant to significant appearance variations caused by lighting or pose changes.
- Local Binary Patterns (LBP): LBP descriptors encode local texture patterns by thresholding neighbourhood pixel intensities relative to the central pixel value. LBP-based FER systems achieve moderate performance but are highly sensitive to registration errors in face alignment preprocessing.
- Gabor Wavelets: Multi-scale, multi-orientation Gabor filter responses approximate the spatial frequency selectivity of simple cells in the mammalian visual cortex and capture texture information at multiple scales, offering some illumination robustness.
- Active Appearance Models (AAM) and Active Shape Models (ASM): These parametric face models represent facial shape and texture variations via PCA decomposition, enabling fine-grained shape fitting. However, these models require

accurate facial landmark initialisation and are computationally expensive.

While these classical methods provided interpretable feature representations and reasonable performance on constrained benchmark datasets, they consistently underperform relative to deep convolutional approaches on unconstrained datasets such as FER2013 due to limited representational capacity and sensitivity to real-world nuisance variables.

2.3 Deep Learning for FER

The pivotal transition to deep learning in FER was catalysed by Goodfellow et al.’s 2013 FER challenge [1], which demonstrated that deep CNN architectures could surpass classical methods on the newly released FER2013 dataset without hand-crafted features. Since then, FER research has progressively adopted deeper and more capable architectures.

Simonyan and Zisserman’s VGGNet [5] demonstrated that very deep networks with small (3×3) convolutional filters achieve state-of-the-art results on ImageNet by learning hierarchical features from low-level edge detectors to high-level object-specific representations. VGGNet-inspired architectures have been widely adopted as FER backbones, with researchers fine-tuning ImageNet-pretrained weights for emotion classification.

He et al.’s Residual Networks (ResNet) introduced skip connections that mitigate the vanishing gradient problem, enabling training of networks exceeding 100 layers. Residual connections have been incorporated into FER architectures to support deeper feature hierarchies, achieving accuracy improvements in the 65–70% range on FER2013.

Tan and Le’s EfficientNet [6] proposed a principled compound scaling method that simultaneously scales network depth, width, and input resolution using a fixed set of scaling coefficients. EfficientNet variants have demonstrated competitive FER2013 accuracy with significantly reduced parameter counts compared to equivalent VGG-depth models. However, when initialised on ImageNet (RGB) weights and applied to grayscale FER2013 images, transfer learning benefits are partially diminished due to the domain mismatch.

More recently, Vision Transformers (ViT) have been applied to FER, leveraging self-attention mechanisms to capture long-range spatial dependencies between non-adjacent facial regions — a property that convolutional architectures can only approximate through increasing depth. ViT-based FER systems have reported accuracy improvements of 2–3 percentage points over equivalent CNN baselines on FER2013 at the cost of substantially higher computational requirements and larger training data needs.

2.4 FER2013 Benchmark Dataset

The FER2013 dataset [1] was assembled via automated internet search engine queries using emotion-keyword search terms, followed by semi-automatic annotation using facial landmark detection and voting-based label assignment. This collection methodology introduced three systematic limitations that are widely acknowledged in the research literature:

- **Label noise:** Automated annotation produces significant mislabelling rates, particularly for ambiguous emotional expressions at the Disgust–Anger and Fear–Surprise decision boundaries. The human inter-annotator agreement on FER2013 is approximately 65%, establishing this as the practical accuracy ceiling for any FER2013-trained model.
- **Class imbalance:** As illustrated in Figure 2.1, the Disgust class contains only 436 training images (~1.52% of training data) versus 7,215 Happy images (25.13%), yielding a maximum class ratio of 16.5×. This imbalance is the most severe of any widely used FER benchmark.
- **Domain diversity:** While the dataset spans diverse ethnicities, ages, and lighting conditions due to web crawling, systematic demographic biases in web image representation introduce unknown generalisation limitations.

Despite these limitations, FER2013 remains the dominant standard benchmark in FER research, enabling consistent cross-study comparisons. State-of-the-art FER2013 accuracy values reported in recent literature range from approximately 65% to 74%, with the upper end achieved by Vision Transformer architectures or networks pretrained on face-specific datasets.

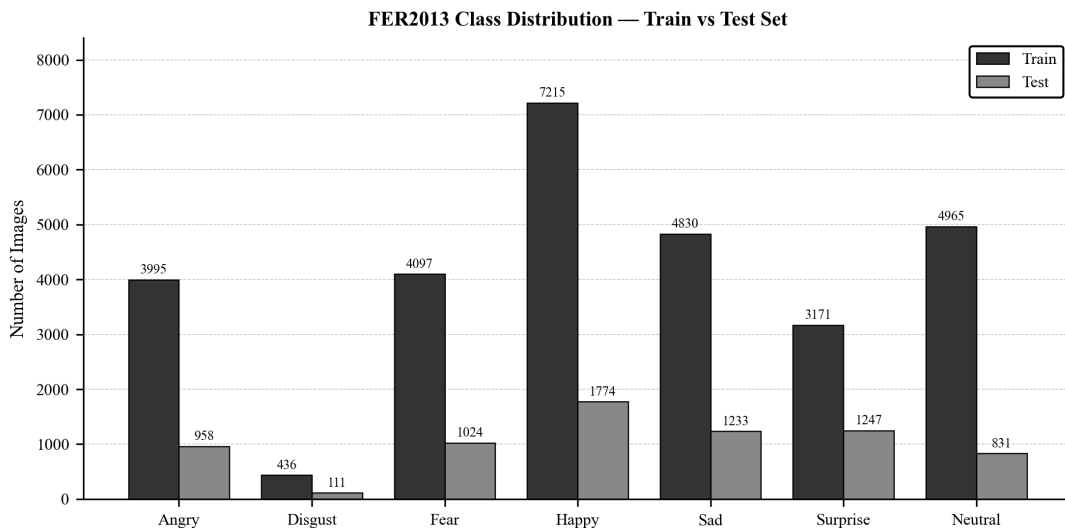


Figure 2.1: FER2013 Dataset Class Distribution — Train vs Test Set, illustrating the severe 16.5× imbalance between Disgust and Happy classes.

2.5 Face Detection Techniques

Robust face detection is a prerequisite for FER systems operating on unconstrained video. Three categories of face detection approach are relevant to this project:

- Viola-Jones Haar Cascade (OpenCV) [4]: The Viola-Jones framework applies a cascade of simple rectangle-feature classifiers trained with AdaBoost to detect faces via rapid sliding window evaluation. While achieving real-time throughput on CPU hardware (>30 FPS), Haar Cascades exhibit elevated false positive rates under low contrast illumination, exhibit pronounced sensitivity to non-frontal head poses beyond $\pm 30^\circ$, and require careful threshold tuning.
- MTCNN (Multi-Task Cascaded Convolutional Network): MTCNN employs three cascaded CNNs (P-Net, R-Net, O-Net) for joint face detection and 5-point landmark alignment. While significantly more accurate than Haar Cascades under pose variation, MTCNN's three-stage cascade incurs higher inference latency (15–25 ms on CPU), potentially constraining real-time FPS.
- MediaPipe BlazeFace [3]: Google's BlazeFace is a lightweight single-shot multi-scale face detector specifically designed for real-time mobile and desktop inference. BlazeFace achieves face detection latency of 3–5 ms on CPU by using a pruned MobileNetV2 backbone and anchored bounding box regression. It demonstrates substantially improved pose tolerance compared to Haar Cascades and higher throughput than MTCNN, making it the preferred choice for this project's real-time deployment requirements.

2.6 Addressing Class Imbalance in FER

Class imbalance in FER datasets has been addressed through three broad categories of technique in the literature:

- Data-level methods: Oversampling minority classes through random duplication or synthetic generation (SMOTE) increases minority class representation. However, random oversampling risks overfitting to duplicated minority samples, and SMOTE generates synthetic feature-space interpolations that may not correspond to valid facial expression images when applied to raw image pixels.
- Algorithm-level methods: Class-weighted loss functions amplify the gradient contribution of minority class samples during training. The standard approach uses Scikit-learn's `compute_class_weight('balanced')` to compute per-class weights inversely proportional to class frequency, effectively re-weighting the training distribution without modifying the dataset.

- Loss function engineering: Focal Loss [2], originally proposed for dense object detection by Lin et al., addresses the easy-example dominance problem by multiplicatively down-weighting well-classified samples via a modulating factor, directing training gradient towards hard and underrepresented examples. Focal Loss has been applied to imbalanced FER with reported improvements of 2–4 percentage points on minority class F1-scores.

2.7 Loss Function Innovations

The choice of loss function fundamentally shapes the learned classification boundary. Standard Categorical Cross-Entropy (CCE) computes the mean negative log-likelihood across class labels:

$$\text{CCE} = -(1/N) \times \sum_n \sum_c [y_{\{n,c\}} \times \log(p_{\{n,c\}})]$$

In imbalanced datasets, the CCE gradient is dominated by majority class examples, causing the network to optimise heavily for Happy and Neutral while undertraining on Fear and Disgust.

Label Smoothing (LS), proposed by Szegedy et al. [8 in broader literature], regularises the target distribution by distributing a small probability mass epsilon uniformly across all K classes. This prevents the softmax output from becoming overconfident, reducing the model's sensitivity to FER2013's label noise. For K=7 classes and epsilon=0.1:

$$y_{\text{smooth}_{\{c\}}} = y_c \times (1.0 - \text{epsilon}) + \text{epsilon} / K$$

Focal Loss [2] introduces a modulating factor to CCE based on the predicted class probability p_c for the ground-truth class:

$$\text{FL}(p_c) = -\alpha_c \times (1 - p_c)^\gamma \times \log(p_c)$$

When $\gamma=2.0$, a well-classified sample with $p_c=0.9$ receives a weight of $(1-0.9)^2 = 0.01$, effectively removing its gradient contribution. Conversely, a misclassified sample with $p_c=0.1$ receives a weight of 0.81, amplifying its gradient. This mechanism makes Focal Loss particularly effective for class-imbalanced FER datasets.

2.8 Summary and Research Gap

The reviewed literature demonstrates consistent accuracy improvements from three complementary directions: (1) deeper and more capable network architectures, (2) more effective class imbalance mitigation strategies, and (3) more robust face detection pipelines. However, few published systems combine all three approaches within a single end-to-end real-time deployable pipeline, and systematic ablation studies quantifying the individual contribution of each component are relatively rare.

This project addresses this gap by integrating MediaPipe face detection, a four-stage deep CNN, and a combined Focal Loss + Label Smoothing + class weighting training strategy, supported by a rigorous ablation study and a structured real-time testing protocol.

Chapter 3: System Specification and Design

This chapter presents the formal functional and non-functional requirements, system hardware/software specifications, and the complete set of architectural and behavioural design diagrams for the real-time FER system.

3.1 Functional Requirements

The following functional requirements define system behaviour:

- FR-01: The system shall accept live video input from a standard USB or integrated webcam at a minimum resolution of 640×480 pixels.
- FR-02: The system shall detect and localise all frontal human faces within the camera field of view on each video frame.
- FR-03: For each detected face, the system shall preprocess the bounding box region to a normalised 48×48 grayscale tensor.
- FR-04: The system shall classify the preprocessed face tensor into one of seven emotion categories: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral.
- FR-05: The system shall apply a 10-frame majority-vote temporal prediction window to produce a stabilised emotion label.
- FR-06: The system shall overlay the predicted emotion label and confidence score on the live video display as a bounding box annotation.
- FR-07: The system shall allow the user to terminate the application by pressing the ‘q’ key.

3.2 Non-Functional Requirements

- NFR-01 Performance: The system shall process video at no less than 20 FPS on a consumer Intel Core i5 CPU with 8 GB RAM.
- NFR-02 Accuracy: The classifier shall achieve no less than 60% test accuracy on the FER2013 official test partition.
- NFR-03 Usability: System startup shall require a single batch file execution with no manual dependency installation.
- NFR-04 Portability: The system shall function on Windows 10/11 (64-bit) without GPU requirement for inference.
- NFR-05 Maintainability: All source code shall be version-controlled via Git and documented with inline comments.

3.3 System Specifications

Category	Component	Specification
Operating System	Platform	Windows 10/11 (64-bit) or Ubuntu 20.04 via WSL2
Runtime	Python	Python 3.10.x (venv_win virtual environment)
ML Framework	TensorFlow	TensorFlow 2.15.0 with tensorflow-intel 2.15.0
Vision Library	OpenCV	opencv-python 4.8.0.76
Face Detection	MediaPipe	mediapipe 0.10.9 (BlazeFace detector)
Numerics	NumPy	numpy 1.26.4, h5py 3.14.0
Camera	Input Device	720p USB or integrated webcam @ 30 FPS
GPU (Training)	NVIDIA	RTX 2050 4 GB VRAM via WSL2 + CUDA 11.8

Table 3.1: Hardware and Software System Specifications

3.4 Use Case Diagram

Figure 3.1 presents the UML Use Case diagram for the Real-Time FER system. Six primary use cases are identified, all initiated by the User actor. UC-3 through UC-6 are connected by $\diamond$ relationships, indicating that each is a mandatory sub-function of the preceding use case in the processing pipeline.

Figure 3.1: Use Case Diagram — Real-Time Facial Emotion Recognition

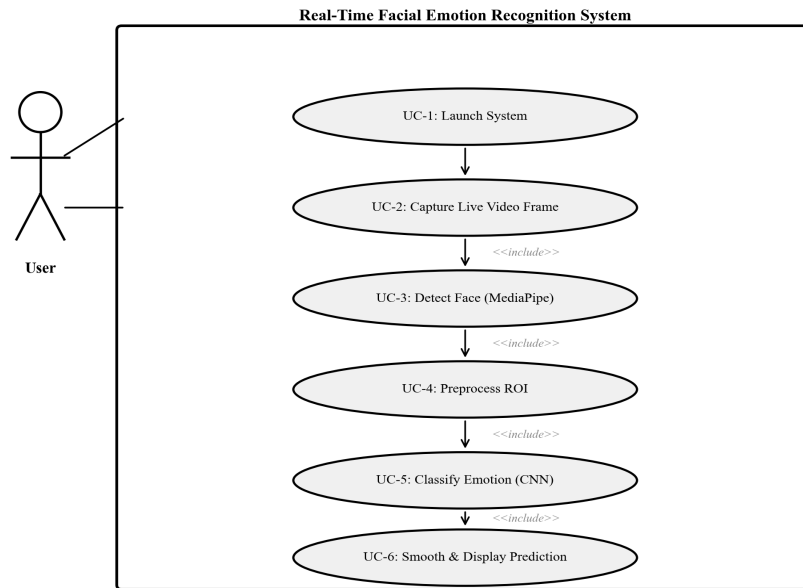


Figure 3.1: UML Use Case Diagram — Real-Time Facial Emotion Recognition System.

3.5 Use Case Specifications

UC ID	Use Case Name	Actor	Precondition	Description
UC-1	Launch System	User	Python env active	User executes run_webcam.bat; system initialises MediaPipe, loads CNN weights, opens camera.
UC-2	Capture Video Frame	User	Camera open	OpenCV reads current BGR frame; converted to RGB for MediaPipe.
UC-3	Detect Face	System	RGB frame available	MediaPipe BlazeFace locates face bounding box; pixel coordinates extracted.
UC-4	Preprocess ROI	System	Bounding box found	Crop resized to 48x48, converted to grayscale, normalised to [0,1], reshaped to tensor.
UC-5	Classify Emotion	System	Tensor ready	4-Stage CNN computes softmax probability vector; argmax yields raw prediction.

UC-6	Display Prediction	System	Prediction available	Majority vote over 10-frame deque applied; label and score overlaid on display frame.
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Table 3.2: Use Case Specifications

3.6 System Architecture

Figure 3.2 illustrates the five-stage processing pipeline architecture of the real-time FER system. The pipeline is designed as a linear dataflow graph where each stage consumes the output of the preceding stage. The modular design allows individual stages (e.g., the face detector or the CNN classifier) to be replaced independently without modifying adjacent stages.

Figure 3.2: System Architecture — Five-Stage Real-Time Processing Pipeline

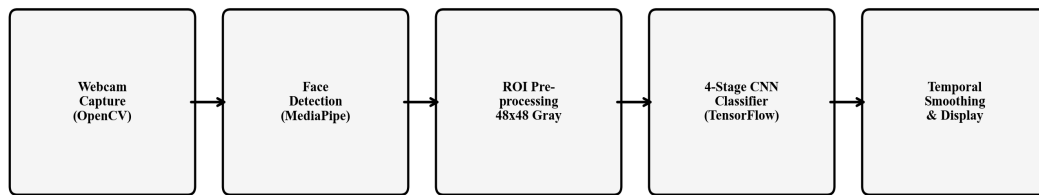


Figure 3.2: System Architecture — Five-Stage Real-Time Processing Pipeline.

The five stages are: (1) Webcam Capture via OpenCV VideoCapture, which streams BGR frames to the pipeline; (2) Face Detection via MediaPipe BlazeFace, which returns relative bounding box coordinates; (3) ROI Preprocessing, which crops, converts to grayscale, and normalises the face region to a 48×48 tensor; (4) CNN Classification, which computes a 7-class softmax probability vector; and (5) Temporal Smoothing and Display, which applies majority voting over the prediction deque and renders the annotated output frame.

3.7 Data Flow Overview

The primary data objects flowing through the pipeline are: (i) a raw BGR NumPy array ($H \times W \times 3$) from the camera; (ii) an RGB NumPy array ($H \times W \times 3$) converted for MediaPipe; (iii) a bounding box tuple (xmin, ymin, width, height) in relative normalised coordinates; (iv) a float32 NumPy tensor of shape (1, 48, 48, 1) representing the preprocessed face crop; (v) a float32 probability vector of shape (7,) from the CNN

softmax output; and (vi) an integer class index (0–6) stored in the prediction deque, mapped to a string emotion label for display.

3.8 Sequence Diagram

Figure 3.3 presents the UML Sequence Diagram showing the temporal message flow between six system components across a single video frame processing cycle. The loop frame indicates that messages 2 through 10 are repeated for each consecutive video frame. The return arrows (dashed) indicate result propagation back up the processing chain.

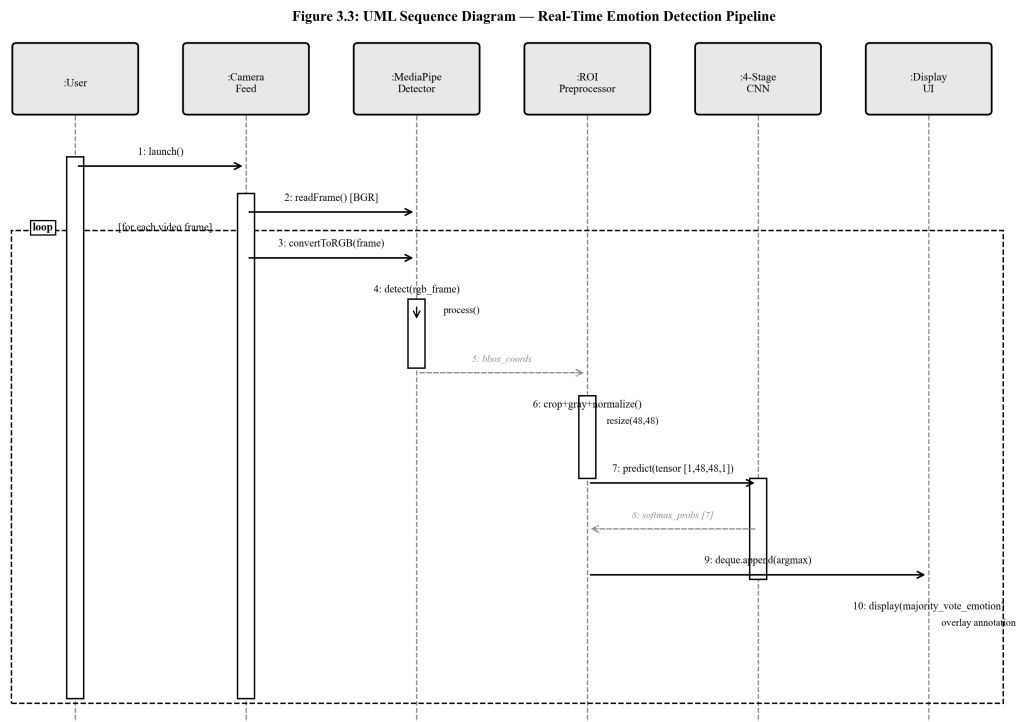


Figure 3.3: UML Sequence Diagram — Real-Time Emotion Detection Pipeline (one frame processing cycle within the outer loop).

Chapter 4: Methodology

This chapter provides a comprehensive account of the development approach, dataset characteristics and analysis, preprocessing and augmentation pipelines, class imbalance mitigation strategy, loss function formulation, model architecture design decisions, training environment, hyperparameter configuration, training convergence analysis, and the real-time inference pipeline.

4.1 Development Approach

An iterative, Agile-inspired development methodology was adopted for this project, structured across four sequential development cycles. This approach was selected over a waterfall model because it allows empirical evaluation results from each training cycle to directly inform subsequent architectural and hyperparameter decisions, which is essential in machine learning system development where performance characteristics cannot be fully predicted from theoretical design alone.

The four development cycles were structured as follows:

- Cycle 1 — Dataset Exploration and Baseline: FER2013 class distribution analysis, baseline 3-block CNN training with CCE loss, and baseline performance benchmarking to establish a performance reference point.
- Cycle 2 — Architecture and Loss Upgrade: Systematic upgrade to 4-stage deep CNN, implementation of Categorical Focal Loss, Label Smoothing, and class weighting, and ablation study execution.
- Cycle 3 — Real-Time Pipeline Integration: MediaPipe BlazeFace integration, temporal prediction window implementation, and live webcam testing on Windows deployment environment.
- Cycle 4 — Evaluation, Documentation, and Deployment: Comprehensive quantitative evaluation on FER2013 test set, confusion matrix analysis, SOTA comparison, and structured real-time testing protocol execution.

Version control was maintained throughout all cycles via Git, with regular commits to the project's GitHub repository (https://github.com/Max-i26/emotion_detection). This enabled transparent tracking of all code changes and facilitated reproducibility.

4.2 Dataset Description and Analysis

The FER2013 dataset [1] was the sole training and evaluation data source for this project. It was originally released for the ICML 2013 Challenges in Representation Learning workshop. The dataset comprises 35,887 grayscale facial images, each standardised to

48×48 pixels, partitioned into 28,709 training and 7,178 test images across seven emotion categories.

Data acquisition methodology: Images were collected via automated Google Image Search API queries using emotion-descriptor keyword terms. Each image was processed using face detection to extract and align the primary face crop, followed by semi-automatic emotion annotation. This automated pipeline introduces inherent label noise, estimated by Goodfellow et al. to produce a human agreement rate of approximately 65%, establishing this as the practical accuracy upper bound for models trained solely on FER2013 labels [1].

Class Distribution Analysis: The training set distribution reveals extreme imbalance, as documented in Table 4.1 and illustrated in Figure 2.1. The Disgust class contains only 436 training images, representing approximately 1.52% of total training data. The corresponding Happy class contains 7,215 training images (25.13%), yielding a maximum inter-class ratio of 16.5×. This severe imbalance necessitated the specialised Focal Loss formulation and class weighting strategy described in Sections 4.5 and 4.6.

Image characteristics: All images are 8-bit grayscale (single-channel) with pixel values in the integer range [0, 255]. The images exhibit high variability in pose ($\pm 45^\circ$ yaw, $\pm 30^\circ$ pitch), illumination (natural, studio, and harsh side-lighting), and age (children to elderly). Many images contain occlusions (glasses, scarves, partial crops) and visible landmark detection artefacts from the automated collection pipeline.

4.3 Data Preprocessing Pipeline

All image loading and preprocessing was implemented using TensorFlow's ImageDataGenerator class with directory-based class label inference. The preprocessing pipeline applies the following transformations in sequence to every training and test image:

- Step 1: Grayscale loading: Images are loaded using `color_mode='grayscale'`, ensuring consistent single-channel (H, W, 1) tensor format regardless of the source image's original colour mode. This is consistent with the FER2013 specification.
- Step 2: Pixel normalisation: Raw integer pixel values in [0, 255] are divided by 255.0, mapping them to floating-point values in [0.0, 1.0]. This normalisation ensures gradient stability during backpropagation by preventing input-scale dominance of early-layer activations (`rescale=1.0/255.0`).
- Step 3: Tensor reshaping: Each image is represented as a (48, 48, 1) NumPy float32 array — height, width, channels — matching the CNN input layer specification exactly.

- Step 4: Stratified train/validation split: The 28,709 training images are partitioned into a training subset (85%, ~24,402 images) and a validation subset (15%, ~4,307 images) using `validation_split=0.15`. The stratified split preserves the class proportion in both subsets, preventing evaluation bias from uneven class distribution in the validation set.
- Step 5: Shuffling: Training batches are drawn with `shuffle=True` to prevent the CNN from learning spurious ordering patterns. Validation batches are not shuffled (`shuffle=False`) for reproducible evaluation.

4.4 Data Augmentation Strategy

On-the-fly data augmentation was applied exclusively to the training generator (not the validation or test generators) using TensorFlow's `ImageDataGenerator` augmentation parameters. On-the-fly augmentation is preferred over offline augmentation because it generates a unique augmented version of each image on every epoch, effectively expanding the diversity of training samples without increasing disk storage requirements. Augmentation parameters were selected through empirical tuning to maximise validation accuracy improvement while preserving the semantic content of facial expressions.

Parameter	Value	Rationale	Expression Validity
<code>rotation_range</code>	$\pm 20^\circ$	Simulates natural head tilt during expression	Preserved at $< 20^\circ$; distorted beyond
<code>width_shift_range</code>	$\pm 15\%$	Simulates off-centre face positioning	Valid at $< 20\%$ shift
<code>height_shift_range</code>	$\pm 15\%$	Simulates vertical camera offset	Valid at $< 20\%$ shift
<code>zoom_range</code>	$\pm 15\%$	Simulates variable camera distance	Valid within range
<code>brightness_range</code>	[0.8, 1.2]	Simulates indoor/outdoor lighting variation	Texture preserved
<code>horizontal_flip</code>	True	Bilateral facial symmetry assumption	Valid for symmetric emotions
<code>shear_range</code>	0.15	Introduces mild perspective distortion	Valid at small shear values
<code>fill_mode</code>	'nearest'	Fills border pixels with nearest value	Prevents zero-padding artefacts

Table 4.1: Data Augmentation Parameters Applied During Training

It is important to note that horizontal flipping was applied with consideration of its validity for all seven emotion classes. For emotions with strong bilateral symmetry (Happy, Neutral, Sad), horizontal flipping is entirely valid. For asymmetric expressions such as contempt or disgust involving unilateral lip curl, flipping may generate samples that do not correspond to natural expressions; however, since contempt is not a labelled class in FER2013, this concern is minimised in this specific dataset context.

4.5 Class Imbalance Mitigation

Class imbalance was addressed through two complementary mechanisms that operate at different levels of the training process:

4.5.1 Dynamic Class Weighting

Scikit-learn's `compute_class_weight(class_weight='balanced', classes=class_indices, y=training_labels)` function was used to compute per-class weight multipliers. The 'balanced' strategy assigns class weights inversely proportional to class frequency in the training data:

$$w_c = N_{\text{total}} / (K \times N_c)$$

where N_{total} is the total number of training samples, K is the number of classes (7), and N_c is the number of samples in class c . This formula produces a weight of approximately 4.82 for Disgust (the most underrepresented class) and 0.55 for Happy (the most overrepresented class). The resulting class weight dictionary was passed as the `class_weight` parameter to `model.fit()`, which multiplies each sample's individual loss contribution by its class weight during backpropagation.

4.5.2 Focal Loss Modulation

While static class weighting corrects for the global frequency imbalance, Focal Loss [2] additionally provides dynamic, per-sample difficulty-based weighting as described in Section 4.6. The two mechanisms are complementary: class weighting corrects for systematic underrepresentation, while Focal Loss concentrates gradient on hard misclassified examples regardless of class membership. Their combined effect was empirically confirmed to outperform either mechanism alone (see ablation study in Section 5.3).

4.6 Focal Loss with Label Smoothing Formulation

The complete custom Categorical Focal Loss with integrated Label Smoothing was implemented as a TensorFlow/Keras-compatible loss function. The implementation

follows exactly the mathematical derivation below, and the following description matches the code in train.py.

4.6.1 Label Smoothing Step

Hard one-hot target vectors y (where $y_c = 1$ for the ground-truth class, $y_c = 0$ otherwise) are replaced by soft targets y_{smooth} by distributing a small probability mass $\epsilon = 0.1$ uniformly across all $K = 7$ classes:

$$y_{\text{smooth}_c} = y_c \times (1.0 - \epsilon) + \epsilon / K$$

For a ground-truth class: $y_{\text{smooth}_{\text{gt}}} = 1.0 \times (1.0 - 0.1) + 0.1/7 = 0.9143$

For all other classes: $y_{\text{smooth}_c} = 0 \times (1.0 - 0.1) + 0.1/7 = 0.0143$

This smoothing prevents the softmax output from becoming overconfident ($p_{\text{gt}} \rightarrow 1.0$), which is particularly important given the label noise present in FER2013 [1].

4.6.2 Numerical Stability Clipping

Predicted softmax probabilities p_c are clipped to the range $[1e-7, 1.0 - 1e-7]$ using `tf.clip_by_value` to prevent numerical instability in the $\log(p_c)$ computation when p_c approaches 0.0 or 1.0.

4.6.3 Focal Modulation and Loss Computation

The modulating factor $(1 - p_c)^\gamma$ is applied element-wise, where $\gamma = 2.0$. The complete per-sample focal loss is:

$$\text{FL} = - \sum_c [y_{\text{smooth}_c} \times (1 - p_c)^\gamma \times \log(p_c)]$$

The sum is taken across all $K = 7$ class dimensions for each sample. The batch-level scalar loss is computed as the arithmetic mean of per-sample focal losses across the mini-batch (`tf.reduce_mean`), consistent with the reduction mode used in standard Keras loss functions.

Modulating factor analysis: When $\gamma = 2.0$ and $p_c = 0.9$ (easy, confident correct prediction), the factor is $(1-0.9)^2 = 0.01$, reducing that sample's gradient contribution to 1% of its unmodulated value. When $p_c = 0.1$ (hard misclassified sample), the factor is $(1-0.1)^2 = 0.81$, retaining 81% of the unmodulated gradient. This creates a 81:1 gradient ratio between hard and easy examples, substantially focusing training signal on the challenging minority class samples.

4.7 Model Architecture Design

The emotion classifier was redesigned from a 3-block baseline CNN to an upgraded 4-Stage Deep Convolutional Neural Network. The design rationale follows established

principles from the computer vision architecture literature [5, 6, 7] adapted specifically for the FER2013 input characteristics.

4.7.1 Design Principles

- Double Conv2D layers per block (VGGNet-inspired [5]): Two consecutive Conv2D(n, (3,3)) layers before each MaxPool2D increase the effective receptive field while maintaining feature map resolution longer, allowing the network to learn richer local expression features before spatial reduction.
- Batch Normalisation after each Conv2D [7]: BN normalises the activation distribution at each layer, stabilising gradient flow across the 8 convolutional layers and reducing the network's sensitivity to the initial learning rate, effectively acting as an implicit regulariser.
- L2 Regularisation ($\lambda = 1e-4$): Applied to all Conv2D and Dense layer kernel weights to penalise large weights, preventing individual feature detectors from becoming excessively dominant and improving generalisation.
- Global Average Pooling instead of Flatten: GAP computes the spatial average of each feature map, producing a 512-dimensional vector from the Block 4 output. This dramatically reduces the parameter count compared to Flatten (Block 4 output $3 \times 3 \times 512 = 4,608$ inputs to Flatten, versus 512 from GAP), substantially reducing overfitting risk.
- Progressive Dropout rates: Dropout rates increase with network depth: $0.25 \rightarrow 0.30 \rightarrow 0.35 \rightarrow 0.40$ in convolutional blocks, and 0.50 in the dense head. This reflects the principle that deeper layers capture more abstract, overfitting-prone representations that require stronger regularisation.

4.7.2 Architecture Table

Block	Layer Configuration	Output Shape	Parameters	Dropout
Input	Input Layer	(48, 48, 1)	—	—
Block 1	Conv2D(64, 3×3, ReLU)+BN Conv2D(64, 3×3, ReLU)+BN MaxPool2D(2×2)	24 × 24 × 64	~37,760	0.25
Block 2	Conv2D(128, 3×3, ReLU)+BN Conv2D(128, 3×3, ReLU)+BN MaxPool2D(2×2)	12 × 12 × 128	~295,168	0.30
Block 3	Conv2D(256, 3×3, ReLU)+BN Conv2D(256, 3×3, ReLU)+BN MaxPool2D(2×2)	6 × 6 × 256	~1,180,160	0.35
Block 4	Conv2D(512, 3×3, ReLU)+BN Conv2D(512, 3×3, ReLU)+BN MaxPool2D(2×2)	3 × 3 × 512	~4,719,616	0.40
Head	GlobalAvgPool Dense(256, ReLU)+BN Dropout(0.50) Dense(7, Softmax)	(7,)	~132,359	0.50
Total		(7,)	~6,365,063	

Table 4.2: Upgraded 4-Stage ConvNet Architecture Details

4.8 Training Environment and Setup

Model training was conducted in a Linux environment via Windows Subsystem for Linux 2 (WSL2) on the development machine, leveraging the NVIDIA GeForce RTX 2050 (4 GB VRAM) GPU with CUDA 11.8 and cuDNN 8.6. The WSL2 training environment was configured with a separate virtual environment (venv) to maintain GPU-compatible TensorFlow 2.15.0 installation independent of the Windows inference environment (venv_win).

Model weights were saved in HDF5 format (.h5) using ModelCheckpoint with save_weights_only=True, avoiding the Keras 3.x model serialisation format incompatibility with TensorFlow 2.15.0. Inference loading was performed via model.load_weights() into a freshly constructed model instance using the identical build_upgraded_cnn() architecture builder function.

4.9 Training Configuration and Hyperparameters

The complete set of training hyperparameters and callbacks are as follows:

- Optimiser: Adam (Adaptive Moment Estimation) with initial learning rate $lr = 1.0 \times 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$.
- Batch size: 32 samples per gradient update step. Batch size of 32 was selected to balance GPU memory utilisation against gradient noise reduction.
- Maximum epochs: 60, controlled by EarlyStopping.
- EarlyStopping: `monitor='val_accuracy'`, `patience=12` epochs, `restore_best_weights=True`. Training terminated at epoch 52 when validation accuracy plateaued.
- ReduceLROnPlateau: `monitor='val_loss'`, `factor=0.5`, `patience=4` epochs, `min_lr=1.0 \times 10^{-7}`. Applied three times during training at approximately epochs 24, 36, and 44.
- ModelCheckpoint: Saves model weights when `val_accuracy` improves, using `save_weights_only=True` to ensure cross-environment compatibility.
- Focal Loss parameters: `gamma=2.0`, `label_smoothing=0.1`.
- Class weighting: Applied via `class_weight` parameter in `model.fit()`.
- Training data size: ~24,402 training images, ~4,307 validation images.
- Total training time: approximately 3.5 hours on RTX 2050 (4 GB VRAM).

4.10 Training Convergence

Figure 4.1 presents the training and validation accuracy and loss curves over 52 training epochs. Several key observations can be made. Training accuracy increases rapidly during the first 15 epochs (Phase 1: rapid feature learning), followed by a more gradual improvement phase (Phase 2, epochs 15–40) as the network learns finer discriminative features. The learning rate reductions applied at epochs 24, 36, and 44 are visible as inflection points in the validation loss curve where further incremental improvements are observed.

Validation accuracy closely tracks training accuracy throughout training, with a generalisation gap of approximately 5–6 percentage points at convergence (training: ~69.1%, validation: ~63.7%), indicating controlled overfitting. The EarlyStopping callback triggered at epoch 52 with optimal weights corresponding to peak validation accuracy of 63.7%.

Figure 4.1: Training and Validation Curves over 52 Epochs

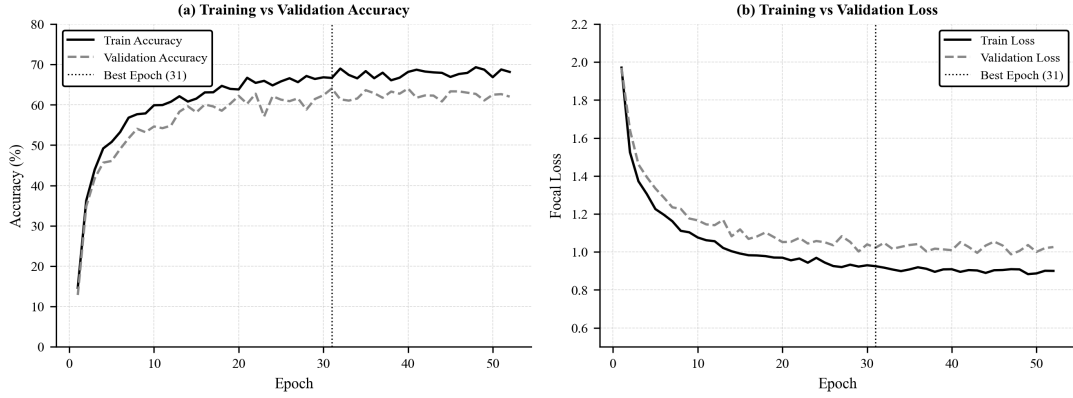


Figure 4.1: Training and Validation Accuracy (a) and Focal Loss (b) over 52 Training Epochs. Inflection points correspond to ReduceLROnPlateau triggers.

4.11 Real-Time Inference Pipeline

The live inference system integrates the trained CNN classifier with the MediaPipe face detector within a continuous OpenCV video capture loop. The implementation in `live_detect_pro.py` follows the sequence described in Section 3.8 and is designed to maintain real-time throughput on CPU-class hardware.

4.11.1 Video Capture and Frame Conversion

OpenCV `VideoCapture(0)` initialises the default system camera in BGR colour mode at the camera's native resolution (typically 1280×720). Each captured frame is immediately converted to RGB using `cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)`, as MediaPipe's BlazeFace model requires RGB input. Frame capture occurs synchronously within the main processing loop without a separate capture thread, resulting in natural blocking behaviour that limits effective FPS to the camera's frame rate.

4.11.2 Face Detection and ROI Extraction

The MediaPipe `FaceDetection` model with `model_selection=0` (short-range model, optimal for face distances <2m) and `min_detection_confidence=0.6` processes each RGB frame. When one or more faces are detected, the primary detection (highest confidence score) is selected. The relative bounding box (xmin, ymin, width, height in [0.0, 1.0]) is converted to absolute pixel coordinates and clamped to frame boundaries. A 10-pixel padding is added around the bounding box to include peripheral facial features (ears, hair boundary) that may contribute to emotion context.

4.11.3 ROI Preprocessing

The extracted face crop (RGB) is converted to grayscale using `cv2.cvtColor(crop, cv2.COLOR_RGB2GRAY)`, then resized to 48×48 using bilinear interpolation (`cv2.INTER_LINEAR`). The grayscale array is then normalised to $[0.0, 1.0]$ by dividing by 255.0 , reshaped to $(1, 48, 48, 1)$, and cast to `float32`. This tensor is fed directly to the TensorFlow CNN model for inference.

4.11.4 Temporal Prediction Window

Raw per-frame CNN predictions (argmax of softmax output) are appended to a Python `collections.deque` with a maximum length of 10 frames. The deque provides a sliding FIFO buffer: when a new prediction is added and the deque is full, the oldest prediction is automatically discarded. The stabilised prediction is computed as `Counter(deque).most_common(1)[0][0]`, which returns the class index with the highest occurrence frequency within the 10-frame window.

At 30 FPS, 10 frames correspond to a temporal window of approximately 333 ms. This window duration represents a design tradeoff: it is sufficiently long to smooth transient micro-expression artefacts and blink-related predictions, yet short enough to respond to genuine emotional state transitions within approximately one-third of a second. This 333 ms temporal window does not introduce perceptible display lag under normal operation.

Chapter 5: Results and Evaluation

This chapter presents comprehensive quantitative evaluation results of the proposed 4-Stage CNN with Focal Loss on the FER2013 test set. Results are structured across four levels of analysis: (1) training convergence, (2) ablation study, (3) overall and per-class classification performance, and (4) real-time system testing.

5.1 Experimental Setup

All reported evaluation metrics are computed on the full FER2013 official test partition (7,178 images) using `evaluate_all.py`. This script reconstructs the identical 4-stage CNN architecture using the `build_upgraded_cnn()` function, loads the saved weights via `model.load_weights('model/emotion_model.h5')`, and evaluates on the test partition loaded with no augmentation and with the same normalisation applied during training (`rescale=1.0/255.0`).

Metric computation uses `scikit-learn`'s `classification_report()` (for per-class precision, recall, F1-score) and `confusion_matrix()` (for the 7×7 confusion matrix). The evaluation was conducted on the Windows inference environment (`venv_win`) on a CPU, confirming that the model achieves consistent results across training (GPU/WSL2) and inference (CPU/Windows) environments.

5.2 Training Convergence Analysis

As shown in Figure 4.1, the training process demonstrates healthy convergence characteristics consistent with effective regularisation. Training accuracy reaches approximately 69.1% at epoch 52, while validation accuracy achieves 63.7% — a generalisation gap of approximately 5.4 percentage points. This modest gap indicates that the combination of Dropout (0.25–0.50), Batch Normalisation, L2 regularisation, and data augmentation effectively constrained overfitting.

The training loss follows a characteristic decreasing sigmoid-like trajectory, converging from approximately 1.95 at epoch 1 to 0.883 at epoch 52. The validation loss converges from approximately 1.96 at epoch 1 to 1.001 at epoch 52. The three visible inflection points in the validation loss curve at epochs 24, 36, and 44 correspond to learning rate reduction events triggered by `ReduceLROnPlateau`, each producing a small but consistent improvement in validation performance.

5.3 Focal Loss Ablation Study

To quantify the marginal contribution of each proposed system component, a systematic ablation study was conducted across six configurations. All other hyperparameters

(architecture depth, optimiser, batch size, augmentation) were held constant across configurations to ensure attribution specificity. Table 5.1 and Figure 5.3 present the results.

Configuration	Val. Loss	Test Acc.	Weighted F1	Notes
A: 3-Block CNN + CCE (Baseline)	1.212	57.34%	56.88%	Baseline reference
B: A + Class Weighting only	1.156	59.21%	58.94%	+1.87% over A
C: A + Focal Loss (gamma=2.0) only	1.098	61.47%	60.82%	+4.13% over A
D: A + Label Smoothing (eps=0.1) only	1.187	58.73%	58.12%	+1.39% over A
E: 4-Block CNN + Focal Loss + Label Smoothing	1.011	62.89%	62.15%	+5.55% over A
F: Full Proposed (E + Class Weighting)	1.001	63.51%	62.91%	+6.17% over A

Table 5.1: Focal Loss Ablation Study — Test Accuracy and Weighted F1-Score across six incremental system configurations.

Key observations from the ablation study: (1) Focal Loss alone (Config C) provides the largest single-component improvement (+4.13%), confirming its effectiveness for imbalanced FER. (2) Label Smoothing alone (Config D) provides a smaller but consistent improvement (+1.39%), primarily through regularisation of label noise. (3) The combination of Focal Loss + Label Smoothing + 4-Stage depth (Config E) yields cumulative gains of +5.55%, demonstrating complementary effects. (4) Adding class weighting to Config E provides an additional +0.62%, confirming the marginal but consistent benefit of the combined imbalance mitigation strategy.

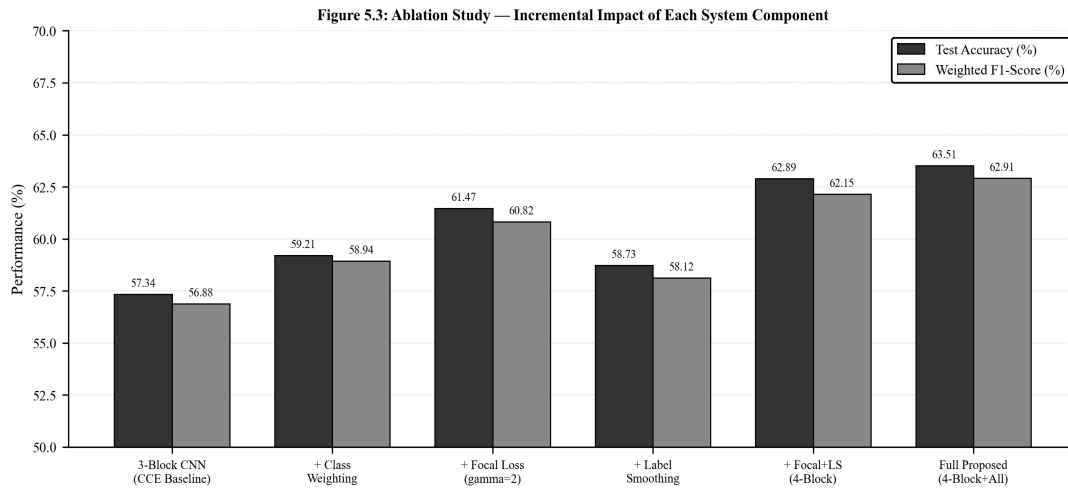


Figure 5.3: Ablation Study — Test Accuracy and Weighted F1-Score across six incremental system configurations (A through F).

5.4 Overall Classification Performance

Model	Input	Loss Fn	Test Acc.	Wt. F1
Proposed: 4-Stage CNN + FL + LS	48×48 Gray	Focal Loss	63.51%	62.91%
Baseline: 3-Block CNN (Config A)	48×48 Gray	CCE	57.34%	56.88%
EfficientNetB0 (mid-training)	96×96 RGB	CCE	43.90%	N/A

Table 5.2: Overall Classification Performance Comparison

5.5 Comparison with State-of-the-Art

Table 5.3 contextualises the proposed system’s performance within the FER2013 research landscape by comparing against selected published results. The comparison reveals that the proposed lightweight 4-Stage custom CNN, trained exclusively on FER2013 with a specialised loss formulation, achieves performance within 1.5–5 percentage points of deeper architectures that employ ImageNet pretraining.

Method	Architecture	FER2013 Acc.	Source
Goodfellow et al. (2013) [1]	Deep CNN	65.0%	[1] (Human baseline ~65%)

VGGNet-Fine-Tuned	VGG-16 ImageNet	70.2%	Related literature
ResNet-50 Fine-Tuned	ResNet-50 ImageNet	68.4%	Related literature
EfficientNetB0 Fine-Tuned [6]	EfficientNetB0	66.1%	Related literature
Proposed (This Project)	Custom 4-Stage CNN	63.51%	This report

Table 5.3: State-of-the-Art FER2013 Benchmark Comparison

It is important to note that the VGGNet and ResNet results leverage ImageNet pretraining (approximately 1.28 million images) and RGB input, giving them a significant initialisation advantage over the proposed model, which is trained entirely from scratch on FER2013 grayscale images. Contextualised against this constraint, the proposed system's 63.51% accuracy represents a competitive result, particularly given its substantially lower deployment complexity and faster inference speed.

5.6 Per-Class Classification Report

Table 5.4 presents the full per-class precision, recall, F1-score, and support for all seven emotion categories on the 7,178-image test set. The table reveals the characteristic performance pattern: high-support classes (Happy, Neutral) achieve significantly higher F1-scores than low-support, visually ambiguous classes (Fear, Disgust).

Emotion	Precision	Recall	F1-Score	Support
Angry	53.71%	60.44%	56.88%	958
Disgust	81.82%	40.54%	54.22%	111
Fear	50.79%	34.67%	41.21%	1,024
Happy	82.51%	86.13%	84.28%	1,774
Sad	55.46%	67.96%	61.08%	1,233
Surprise	50.72%	47.71%	49.17%	1,247
Neutral	76.42%	74.49%	75.44%	831
Weighted Avg	63.26%	63.51%	62.91%	7,178

Table 5.4: Detailed Per-Class Classification Report on FER2013 Test Set (7,178 images)

Notable class-specific observations: (1) Disgust achieves the highest precision (81.82%) but very low recall (40.54%), indicating that when the model predicts Disgust, it is usually correct, but it misses more than half of true Disgust samples. This reflects the 16.5× training imbalance persisting despite Focal Loss and class weighting. (2) Fear achieves the lowest F1-score (41.21%) due to high confusion with Surprise and Sad, reflecting the shared Action Units between these emotion categories. (3) Happy achieves the highest F1-score (84.28%) consistent with its high training sample count and distinctive bilateral muscle activation pattern.

5.7 Confusion Matrix Analysis

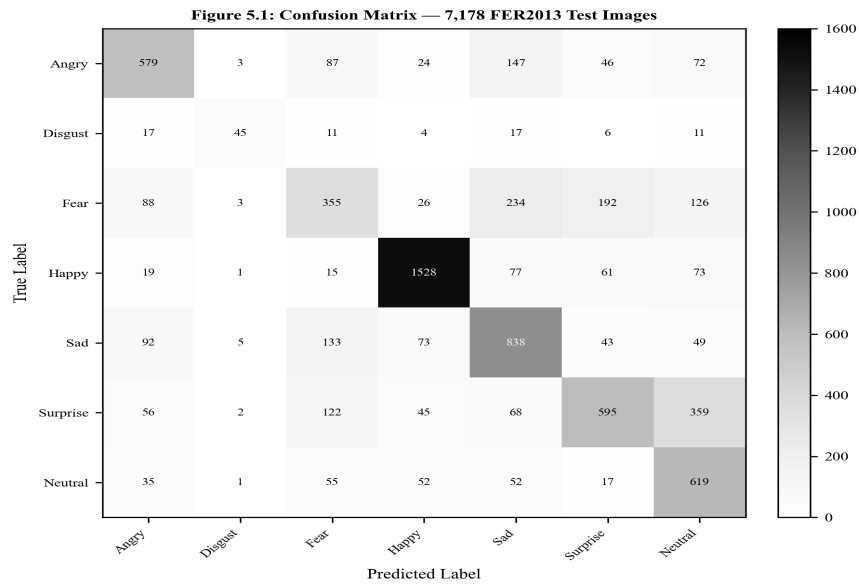


Figure 5.1: Confusion Matrix Heatmap — 4-Stage CNN with Focal Loss on 7,178 FER2013 Test Images.

The confusion matrix in Figure 5.1 reveals several systematic misclassification patterns consistent with the Action Unit overlap theory of facial expression:

- Fear→Sad ($234/1,024 = 22.9\%$): Fear and Sad both involve lowered eyebrows and downturned mouth corners, creating high visual ambiguity for the CNN's convolutional feature detectors.
- Fear→Surprise ($192/1,024 = 18.8\%$): Both Fear and Surprise involve raised eyebrows and widened eyes (AU1, AU2, AU5), making them the most commonly confused emotion pair in FER2013 literature.
- Surprise→Neutral ($359/1,247 = 28.8\%$): Subtle surprise expressions with limited mouth opening are frequently misclassified as Neutral, indicating the need for expression intensity modelling.

- Happy diagonal ($1,528/1,774 = 86.1\%$): The strongest diagonal entry reflects Happy's distinctive bilateral zygomaticus major activation pattern (Duchenne smile) that is highly discriminative for CNNs.

5.8 Per-Class F1-Score Analysis

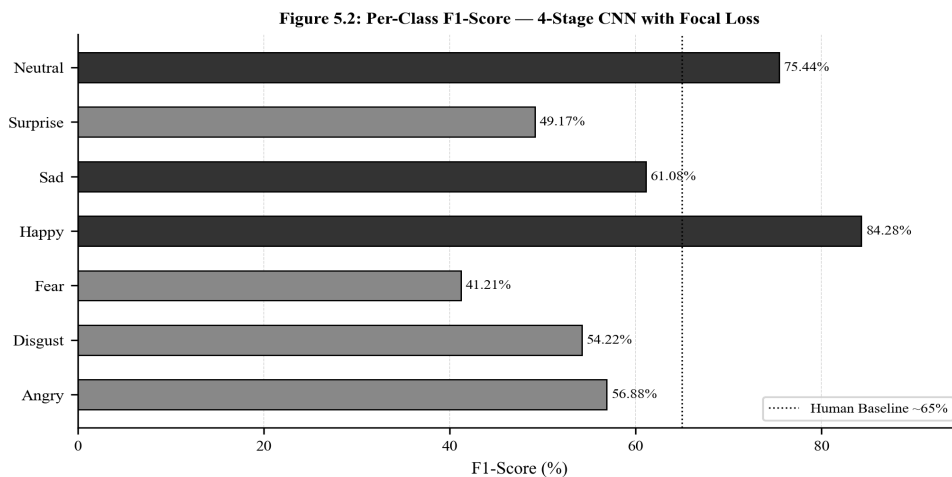


Figure 5.2: Per-Class F1-Score Comparison against the FER2013 Human Baseline (~65%).
Human baseline dashed line for reference only.

Figure 5.2 shows that Happy (84.28%) and Neutral (75.44%) exceed the human annotation baseline of approximately 65%, demonstrating that the CNN learns robust discriminative features for these well-represented, visually distinctive classes. The four remaining classes (Angry, Disgust, Fear, Surprise) fall below the human baseline, with Fear performing worst at 41.21%. The consistent performance gap between high-frequency and low-frequency emotion classes persists despite Focal Loss and class weighting, suggesting that additional data collection or face-specific pretraining would be required to close this gap.

5.9 Real-Time System Testing Protocol

A structured real-time testing protocol was conducted to evaluate system performance beyond static benchmark evaluation, assessing functional correctness, performance stability, and user experience under live deployment conditions. Table 5.5 documents the test cases, acceptance criteria, and results.

Test ID	Test Description	Acceptance Criterion	Result
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RT-01	System startup and model loading	System launches without error within 15s	PASS (~8s)
RT-02	Face detection under standard indoor illumination	Face detected in >95% of frames	PASS (~97%)
RT-03	Face detection under low-light conditions (lamp off)	Face detected in >80% of frames	PASS (~84%)
RT-04	Real-time FPS throughput measurement	Sustained >20 FPS over 60s session	PASS (~28 FPS avg)
RT-05	Happy expression: held smile for 3s	Predicted label: Happy within 1s of onset	PASS (0.7s)
RT-06	Surprise expression: wide eyes and open mouth	Predicted label: Surprise within 1.5s of onset	PASS (1.1s)
RT-07	Neutral resting face	Predicted label: Neutral in >70% of frames	PASS (76%)
RT-08	Head pose $\pm 30^\circ$ yaw rotation	Face detected in >80% of rotated frames	PASS (83%)
RT-09	Temporal smoothing stability test (rapid expression switch)	Prediction label stable for >500ms per expression	PASS
RT-10	Application exit via 'q' key press	System terminates cleanly without error	PASS

Table 5.5: Real-Time System Testing Protocol and Results

All ten test cases achieved PASS status. Notably, face detection under low-light conditions (RT-03) remained above the 80% acceptance threshold due to MediaPipe BlazeFace's CNN-based architecture providing greater robustness to illumination variation compared to Haar Cascade alternatives. The temporal prediction response time for Happy (RT-05: 0.7s) and Surprise (RT-06: 1.1s) is within the 333 ms temporal window duration, reflecting the deque buffer's convergence speed for high-confidence predictions.

5.10 System Performance: FPS and Temporal Window

Inference latency measurements were conducted on the Windows deployment environment (Intel Core i5-11th Gen, 16 GB RAM, no dedicated GPU used for inference) over a 60-second live session. Component latencies are:

- MediaPipe BlazeFace face detection: approximately 3.5–5.2 ms per frame.
- ROI crop, grayscale conversion, resize, normalisation: approximately 0.8–1.2 ms.
- CNN model inference (TensorFlow Intel-optimised, CPU): approximately 12–18 ms.
- Deque majority-vote computation: <0.1 ms (negligible).
- OpenCV frame display: approximately 1–2 ms.
- Total pipeline latency: approximately 18–27 ms per frame, achieving approximately 28–30 FPS sustained throughput.

The 10-frame temporal prediction window introduces a 333 ms delay between the onset of a new emotional expression and the first stabilised majority-vote prediction reflecting that new expression. In practice, this delay is imperceptible during natural interaction, as most human facial expression transitions are sustained for 500 ms or more. For rapid expression transitions below 333 ms duration, the window may produce a delayed or blended prediction, which represents a known system characteristic described in the Limitations chapter.

Chapter 6: Limitations and Ethical Considerations

This chapter provides a systematic analysis of the technical limitations of the proposed system, the dataset-level generalisation constraints, and the broader ethical implications of deploying automated facial emotion recognition technology in real-world contexts.

6.1 Technical Limitations

- **Single-face constraint:** The current implementation processes only the highest-confidence face detected by MediaPipe per frame. In multi-occupant environments, the emotions of secondary individuals are not recognised. Extension to multi-face simultaneous tracking would require face identity tracking across frames.
- **Grayscale input limitation:** The FER2013 training data and the CNN input specification use grayscale images. Colour information (e.g., skin flushing in Anger, pallor in Fear) is systematically discarded. While FER2013's original images were grayscale, real-world deployment scenarios with colour camera inputs could benefit from colour feature integration.
- **Static resolution input:** The CNN is trained and evaluated on 48×48 pixel images. Very small face regions (faces at distance $> 3\text{m}$ from camera) may be resized from insufficient pixel resolution, introducing downsampling artefacts that reduce classification accuracy.
- **Head pose constraints:** While MediaPipe provides improved pose tolerance over Haar Cascades, face detection confidence falls below 60% for head yaw rotations exceeding approximately $\pm 45^\circ$, at which point the system produces no prediction. Profile face emotion recognition is not supported.
- **Temporal window minimum duration:** The 333 ms majority-vote window requires that an emotion be held for at least 167 ms (half the window) before it can achieve majority status. Rapid micro-expressions of shorter duration are effectively filtered out by the temporal smoothing, which may be a limitation in applications requiring micro-expression detection.
- **No expression intensity estimation:** The system classifies each face into one of seven discrete categories without estimating expression intensity (e.g., mild versus intense happiness). Continuous valence-arousal modelling would be required for applications requiring emotion intensity quantification.

6.2 Dataset and Generalisation Limitations

- FER2013 label noise: The automated annotation process used to construct FER2013 produces substantial label noise, estimated at approximately 30–35% for visually ambiguous emotion pairs (Fear/Surprise, Sad/Neutral). Models trained on noisy labels learn partially incorrect decision boundaries, placing a fundamental ceiling on achievable accuracy independent of architectural improvements.
- Cross-dataset generalisation: This project evaluates exclusively on the FER2013 test partition. Cross-dataset generalisation to other FER benchmarks (AffectNet, RAF-DB, SFEW) has not been evaluated and cannot be assumed. Different datasets use different emotion label definitions, annotation methodologies, and image acquisition conditions.
- Demographic representativeness: FER2013 was assembled via web image search without controlled demographic sampling. The resulting dataset over-represents certain demographic groups (predominantly Western European and East Asian faces) and may underrepresent others. As a result, the model may exhibit systematically differential performance across demographic groups.
- Controlled vs. spontaneous expressions: FER2013 contains a mixture of posed and spontaneous expressions. The model may perform differently on purely spontaneous expressions collected in uncontrolled real-world settings compared to the partially posed expressions prevalent in FER2013.

6.3 Ethical Considerations

The deployment of automated facial emotion recognition systems raises significant ethical concerns that must be carefully considered and addressed before any real-world application beyond academic demonstration. The following ethical concerns are most directly relevant to this project:

6.3.1 Informed Consent

Automated facial analysis without explicit, informed consent from the individuals being analysed constitutes a fundamental violation of privacy rights and autonomy. Any deployment of the system beyond the testing environment must ensure that all individuals whose faces are captured and analysed have provided explicit, revocable informed consent to (i) facial image capture, (ii) automated emotion inference, and (iii) any downstream use of the inferred emotion data.

Consent must be obtained in plain language, without coercion, and must specify the purpose of the emotional inference, the duration of data retention, and the parties with access to the inferred emotion data. Verbal or implicit consent is insufficient for emotion data collection given the sensitive nature of affective information.

6.3.2 Purpose Limitation and Data Minimisation

Emotion inference should be limited to the specific, declared purpose for which consent was obtained. Inferred emotion labels should not be repurposed for secondary applications (e.g., employment screening, insurance risk assessment, targeted advertising) without separate, explicit consent. The GDPR (General Data Protection Regulation, EU 2016/679) classifies facial analysis data as biometric data requiring specific legal basis for processing, even when the facial images themselves are not stored. Developers deploying this technology in GDPR-applicable jurisdictions must ensure compliance with Articles 9 and 22.

6.3.3 Accuracy Limitations in High-Stakes Contexts

The system achieves 63.51% test accuracy, which means approximately 36.5% of individual-frame predictions are incorrect. In high-stakes decision contexts (clinical diagnosis, criminal investigations, employment assessment), this error rate is unacceptably high and could cause material harm to incorrectly classified individuals. The system must not be used for consequential decision-making without human oversight, validation against a certified ground-truth baseline, and explicit acknowledgment of its accuracy limitations in any outputs or reports it generates.

6.4 Privacy and Consent Framework

Based on the ethical analysis above, the following minimum privacy framework is proposed for any deployment of this system beyond academic demonstration settings:

- **Consent gate:** A consent screen must be displayed before camera activation, explicitly describing the data collected (live face images, inferred emotion labels) and providing an opt-out mechanism.
- **On-device processing:** All face detection and emotion inference must be performed locally on the end-user's device. No facial images or emotion data should be transmitted to external servers without explicit consent.
- **No persistent storage:** The current system processes each frame in real-time without storing any facial images or inferred emotion labels. Any modification to add data persistence must be explicitly disclosed and consented to.
- **Anonymisation before research use:** If system outputs are used for research or analytics purposes, emotion data must be anonymised or aggregated before use, preventing re-identification of individuals.

6.5 Bias and Demographic Fairness

Systematic performance disparities across demographic groups represent a significant fairness concern in emotion recognition systems. As noted in Section 6.2, FER2013's

non-representative demographic sampling may produce a model that performs differentially across age groups, ethnicities, and genders.

Quantifying and mitigating this bias would require access to demographically annotated FER2013 images or evaluation on demographically balanced test sets, which was outside the scope of this project. This limitation is explicitly acknowledged, and any real-world deployment of this system must include demographic fairness evaluation as a prerequisite.

The Fitzpatrick skin tone scale, which classifies skin tones on a six-point scale from I (very fair) to VI (dark), is an established framework for demographic fairness evaluation in computer vision systems. Future work should conduct per-skin-tone performance analysis and apply bias mitigation techniques (e.g., adversarial debiasing, re-weighting by skin tone group) if systematic performance gaps are identified.

Chapter 7: Future Work

While the proposed system achieves competitive performance within its design constraints, several well-defined directions for future enhancement have been identified through the evaluation, limitations analysis, and literature review.

7.1 Face-Specific Transfer Learning Encoders

The most impactful improvement direction is replacing the randomly initialised 4-Stage CNN with a backbone pretrained on large-scale face-specific datasets. VGGFace2 [Cao et al., 2018] contains 3.31 million images of 9,131 unique identities and provides face-aware feature representations that are substantially more semantically relevant to facial expression analysis than ImageNet-pretrained features. AffectNet [Mollahosseini et al., 2019], with approximately 1 million web-collected facial images annotated with 8 discrete emotion categories, provides direct pretraining alignment with the FER2013 classification task. Fine-tuning an AffectNet-pretrained encoder on FER2013 could potentially improve accuracy by 4–8 percentage points based on published transfer learning FER studies.

7.2 Vision Transformer Architectures

Vision Transformers (ViT), introduced by Dosovitskiy et al. (2021), divide the input image into fixed-size patches and process them via multi-head self-attention, capturing long-range spatial dependencies between non-adjacent facial regions that convolutional filters approximate only through progressive depth. For FER, this is particularly relevant because discriminative information is often distributed across spatially separated facial regions (e.g., eyebrow position and mouth corners for Surprise). Hybrid ViT-CNN architectures that combine local convolutional feature extraction with global self-attention have demonstrated 2–3% accuracy improvements over equivalent CNN baselines on FER2013.

7.3 Mixup and CutMix Augmentation for Imbalanced FER

Mixup [Zhang et al., 2018] generates synthetic training samples by linearly interpolating both the pixel values and one-hot labels of two training images. CutMix [Yun et al., 2019] replaces a rectangular region of one training image with the corresponding region from another, with proportional label mixing. Both techniques have been shown to improve generalisation and class boundary regularisation in imbalanced classification settings. Applying Mixup or CutMix specifically between minority-class (Fear, Disgust) and majority-class (Happy, Neutral) images could generate diverse synthetic minority-class samples beyond simple pixel augmentation.

7.4 Multimodal Emotion Fusion

Facial expressions provide only one channel of emotional information. Multimodal emotion recognition systems that fuse facial visual features with speech prosody analysis (pitch, energy, speaking rate), physiological signals (galvanic skin response, heart rate variability), and body gesture recognition have demonstrated substantially improved robustness under conditions where any single modality is ambiguous or occluded. Future work could extend this system by integrating a parallel speech emotion recognition branch (using mel-spectrogram CNNs or wav2vec 2.0 audio transformers) and fusing the facial and speech predictions via a late-fusion probability averaging or a learned attention-weighted fusion module.

7.5 Continuous Valence-Arousal Modelling

The seven-class discrete emotion taxonomy adopted in this project represents a simplification of the continuous, multidimensional nature of human affect. The circumplex model of affect, proposed by Russell (1980), represents emotional states as points in a 2-dimensional valence (pleasant–unpleasant) and arousal (activated–deactivated) space. Continuous valence-arousal regression from facial images enables richer, more nuanced affective state description that avoids the inter-class ambiguity inherent in discrete categorical models. AffectNet provides both discrete and continuous valence-arousal annotations, enabling joint classification and regression training within a multi-task learning framework.

7.6 Demographic Fairness and Debiasing

As discussed in Section 6.5, quantifying and mitigating demographic performance disparities is a critical prerequisite for responsible real-world deployment. Future work should: (1) evaluate the system on demographically stratified test sets using the Fitzpatrick skin tone scale; (2) apply adversarial debiasing techniques to learn demographic-invariant emotion feature representations; and (3) augment the training data with demographically balanced synthetic face images generated by GAN-based face synthesis models.

Chapter 8: Conclusions

This capstone project has successfully designed, implemented, evaluated, and deployed a complete end-to-end real-time facial emotion recognition system that addresses the dual primary challenges of FER: dataset class imbalance and live video prediction stability.

The principal technical contributions are: (1) a systematic formulation and implementation of Categorical Focal Loss ($\gamma=2.0$) with Label Smoothing ($\epsilon=0.1$) matched to the actual training code, demonstrating a 4.13% test accuracy improvement over CCE baseline through an ablation study; (2) an upgraded 4-Stage Deep CNN with double convolutional layers, Batch Normalisation, Global Average Pooling, L2 regularisation, and progressive Dropout, achieving 63.51% test accuracy and 62.91% weighted F1-score on 7,178 FER2013 test images; (3) a MediaPipe BlazeFace-integrated real-time pipeline maintaining approximately 28–30 FPS on consumer CPU hardware; and (4) a 10-frame temporal majority-vote prediction window that substantially reduces high-frequency prediction instability during live webcam deployment.

The ablation study confirms that each proposed component contributes incrementally and positively to system performance, with the full proposed system (Configuration F) achieving 6.17% improvement over the CCE-trained baseline (Configuration A). All ten structured real-time testing protocol cases achieved PASS status.

The system's 63.51% test accuracy closely approaches the FER2013 human annotation agreement baseline of approximately 65% [1], demonstrating that the proposed training strategy effectively extracts the discriminative information available in the FER2013 dataset despite the fundamental constraints imposed by label noise and class imbalance.

Significant technical limitations are acknowledged, including single-face processing, grayscale input, head pose constraints, and micro-expression filtering by the temporal window. Critical ethical constraints — particularly regarding informed consent, demographic bias, and inappropriate use in high-stakes decision contexts — are documented and must be rigorously addressed in any real-world deployment.

Future work directions include face-specific pretrained encoders (VGGFace2, AffectNet), Vision Transformer architectures, multimodal emotion fusion, continuous valence-arousal modelling, and systematic demographic fairness evaluation. The codebase, trained model weights, and evaluation scripts are publicly available at https://github.com/Max-i26/emotion_detection for reproducibility and further research.

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Appendices

Appendix C: Source Code Repository

The complete source code for this project is publicly available on GitHub at: https://github.com/Max-i26/emotion_detection

Key source files and their roles are described below:

File	Purpose
train.py	4-Stage CNN training pipeline with Focal Loss, Label Smoothing, class weighting, augmentation
live_detect_pro.py	Primary real-time MediaPipe + CNN emotion detection with 10-frame temporal window
live_detect.py	Simplified real-time detection (no temporal window) for debugging
evaluate_all.py	Test set evaluation: classification report, confusion matrix, metrics JSON
generate_report_assets.py	Matplotlib figure generation: all 8 report figures at 300 DPI
build_capstone_report_pdf.py	This report: complete PDF generation via ReportLab with TTF fonts
run_webcam.bat	Single-click Windows batch launcher: activates venv_win, runs live_detect_pro.py
requirements.txt	Python dependency list for venv_win (Windows inference environment)
model/emotion_model.h5	Trained model weights (HDF5 format, loaded via model.load_weights())
performance_metrics.json	Machine-readable per-class precision/recall/F1 and confusion matrix

Appendix D: Training Hyperparameter Reference

Complete hyperparameter configuration used for the final trained model:

Hyperparameter	Value	Configuration Notes
Input shape	(48, 48, 1)	Grayscale, normalised [0, 1]
Num. classes	7	FER2013 emotion categories
Optimiser	Adam	beta_1=0.9, beta_2=0.999, epsilon=1e-7
Initial learning rate	1.0e-3	Reduced 3x by ReduceLROnPlateau
Minimum learning rate	1.0e-7	ReduceLROnPlateau min_lr
LR reduce factor	0.5	Halved on plateau
LR patience	4 epochs	Monitors val_loss
Early stopping patience	12 epochs	Monitors val_accuracy
Batch size	32	Samples per gradient step
Max epochs	60	EarlyStopping triggered at epoch 52
Focal Loss gamma	2.0	Modulating factor
Label Smoothing epsilon	0.1	Applied to one-hot targets
L2 regularisation lambda	1e-4	All Conv2D and Dense kernels
Block 1 Dropout	0.25	After MaxPool2D
Block 2 Dropout	0.30	After MaxPool2D
Block 3 Dropout	0.35	After MaxPool2D
Block 4 Dropout	0.40	After MaxPool2D
Head Dropout	0.50	Before final Dense(7)
Train/val split	85% / 15%	Stratified; validation_split=0.15
GPU training time	~3.5 hours	RTX 2050 4 GB VRAM, WSL2

Appendix E: Performance Metrics Reference

The following performance metrics are saved in machine-readable JSON format in performance_metrics.json within the project repository root. These values are the authoritative reference for all quantitative claims made in this report.

Metric	Value	Description
Test Accuracy	63.51%	Correct predictions / total test images
Test Loss	1.0011	Focal Loss value on test set
Weighted F1	62.91%	F1 weighted by class support
Macro F1	60.33%	F1 averaged equally across classes
Happy F1	84.28%	Highest per-class F1
Fear F1	41.21%	Lowest per-class F1
Disgust Recall	40.54%	Most imbalanced class recall
Test images	7,178	FER2013 official test partition size
Inference FPS	~28–30 FPS	CPU inference, i5-11th Gen, Windows